

What we can assess at 30 m, and what we cannot.

Intervention	Expected satellite signature	Detectable at 30 m?	Ceiling
Plantation / afforestation	sustained NDVI rise over the block, multi-year trajectory	Yes, clearly, above 1 ha	L4
Water-body renovation	water persistence and maximum extent versus pre	Yes — JRC gives a free baseline	L4
Check dam	MNDWI rise at site; persistence months/yr; rabi NDVI in command	Often — impoundment 1,000–10,000 m ²	L4
Percolation tank	tank visible; recharge effect is indirect only	Tank yes, recharge no	L3
Farm pond	marginal MNDWI; rabi NDVI in the owner’s field	Usually not — 400–2,500 m ²	L2–L3 cluster
Contour trench / gully plug	block-level NDVI and bare-soil change	Individually no; block yes	L2–L3 cluster
Dug well / borewell	no reliable optical signature	No	L1 — existence only
Livestock / livelihood	none	No	Out of scope — stated

Publishing the list of things the system cannot assess is what makes the rest of the table believable. The last two rows are deliberate.

Scope discipline: what we deliberately did not build.

NOT BUILT — ON PURPOSE

- A custom capture app — DRISHTI already exists nationally
- A single “Watershed Health Score / 100” — arbitrary weights are an attack surface
- Object detection of structures from 30 m imagery — physically impossible
- Super-resolution of satellite chips — hallucination in an evidence system
- A hydrological simulation model — unverifiable in five weeks
- Any live government-login dependency in the demo
- WebSockets, GraphQL, Kubernetes, microservices, blockchain
- Any model trained from scratch

THE FIVE RISKS WE ACTUALLY LOSE TO

Sub-pixel structures

detectability gate + cluster escalation; designed for, not discovered

Monsoon cloud

rabi and summer carry the analysis; AOI-specific cloud accounting

A false negative harming a beneficiary

N3 needs two independent families and an excluded alternative; language locked to “requires verification”

No access to SRISHTI data

built to their schema behind a driver interface; fully functional on open data

Scope creep

a written ten-item do-not-build list and a Friday demo-or-cut checkpoint